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Yang et al.

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(54) **METHOD OF MANUFACTURING
INTEGRATED CIRCUIT**

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CPC **G03F 7/70033** (2013.01); **G03F 7/702**
(2013.01)

(58) **Field of Classification Search**
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See application file for complete search history.

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Primary Examiner — David E Smith

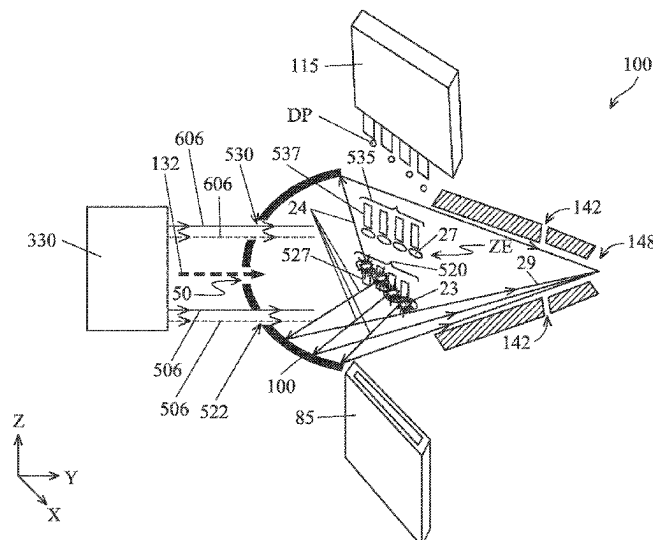
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(57) **ABSTRACT**

A method for generating an extreme ultraviolet (EUV)
radiation includes simultaneously irradiating two or more
target droplets with laser light in an EUV radiation source
apparatus to produce EUV radiation and collecting and
directing the EUV radiation produced from the two or more
target droplet by an imaging mirror.

20 Claims, 11 Drawing Sheets



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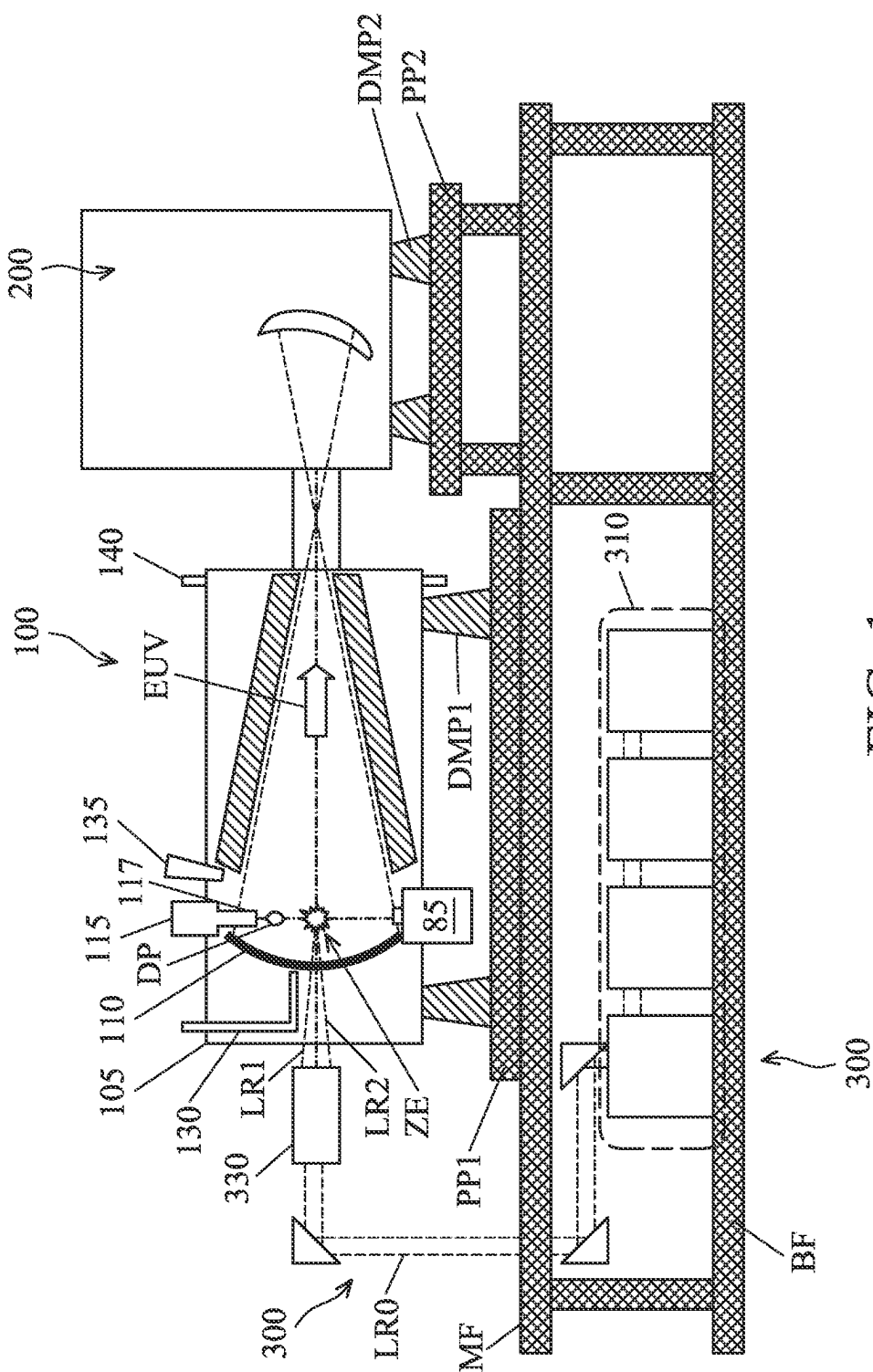
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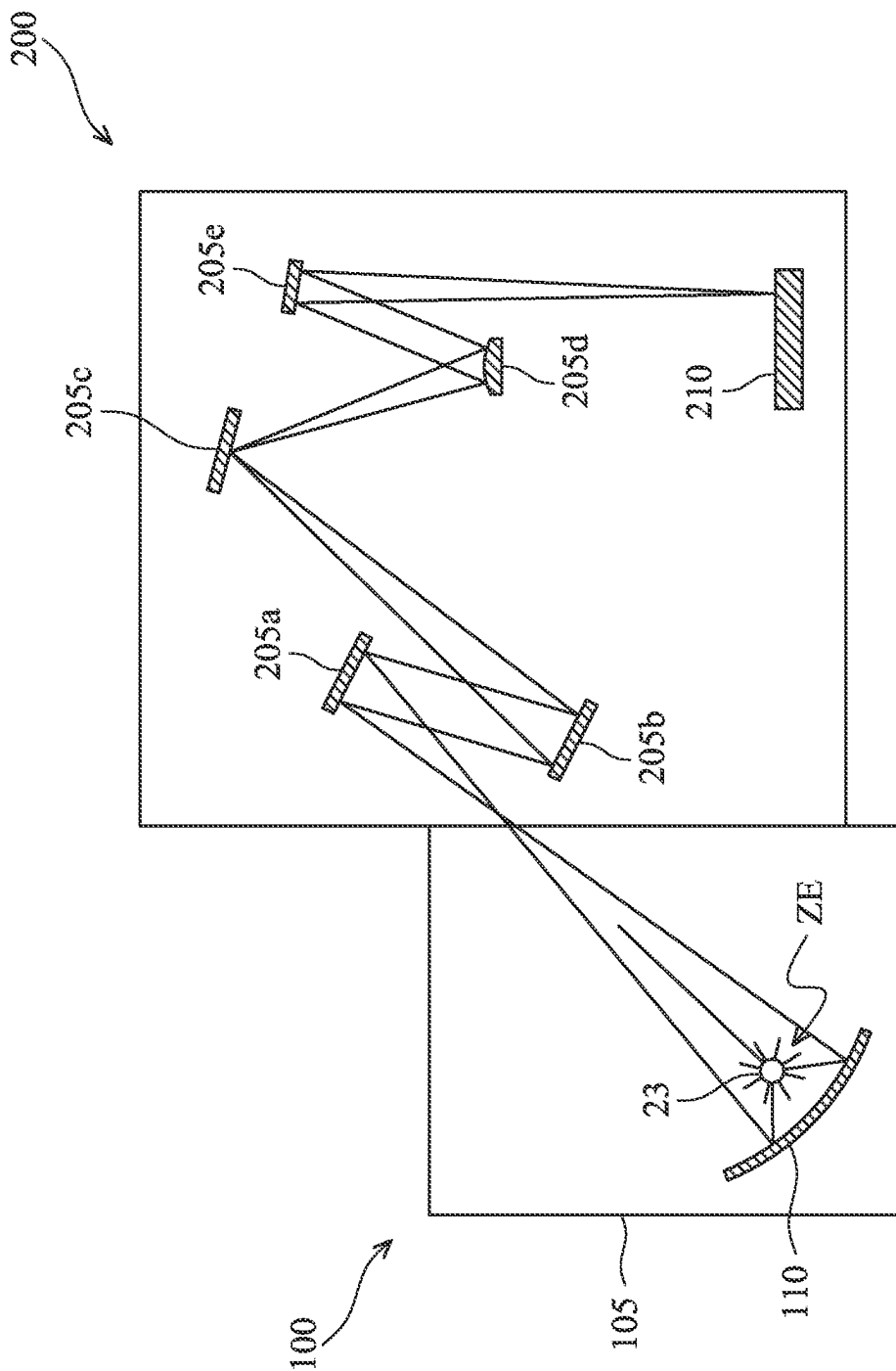


FIG. 2

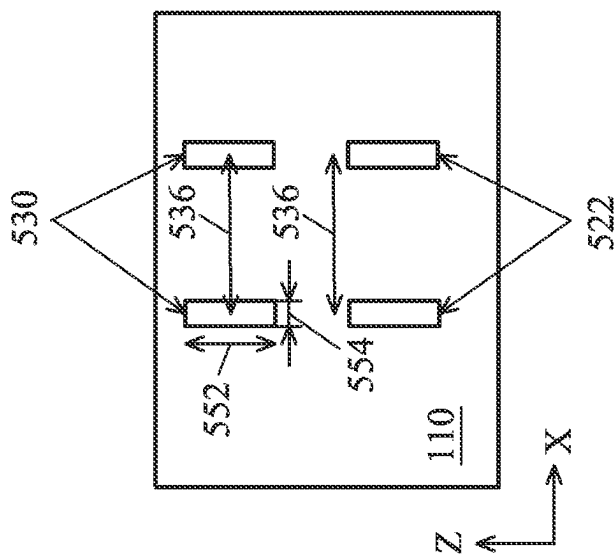


FIG. 3B

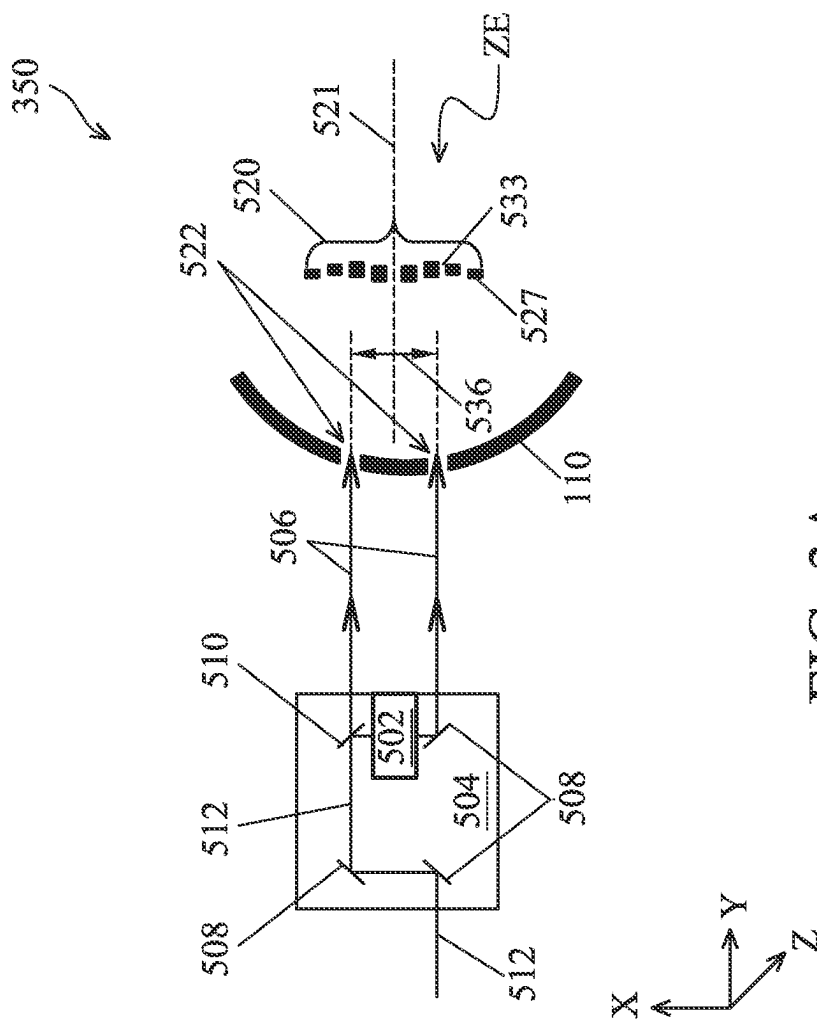


FIG. 3A

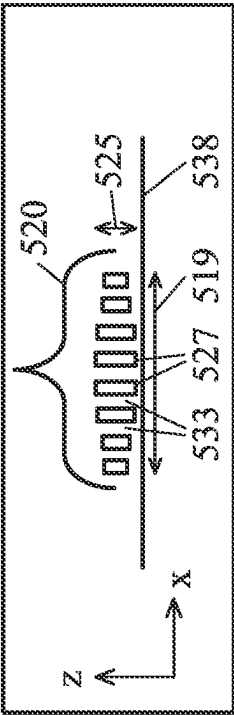


FIG. 5

400

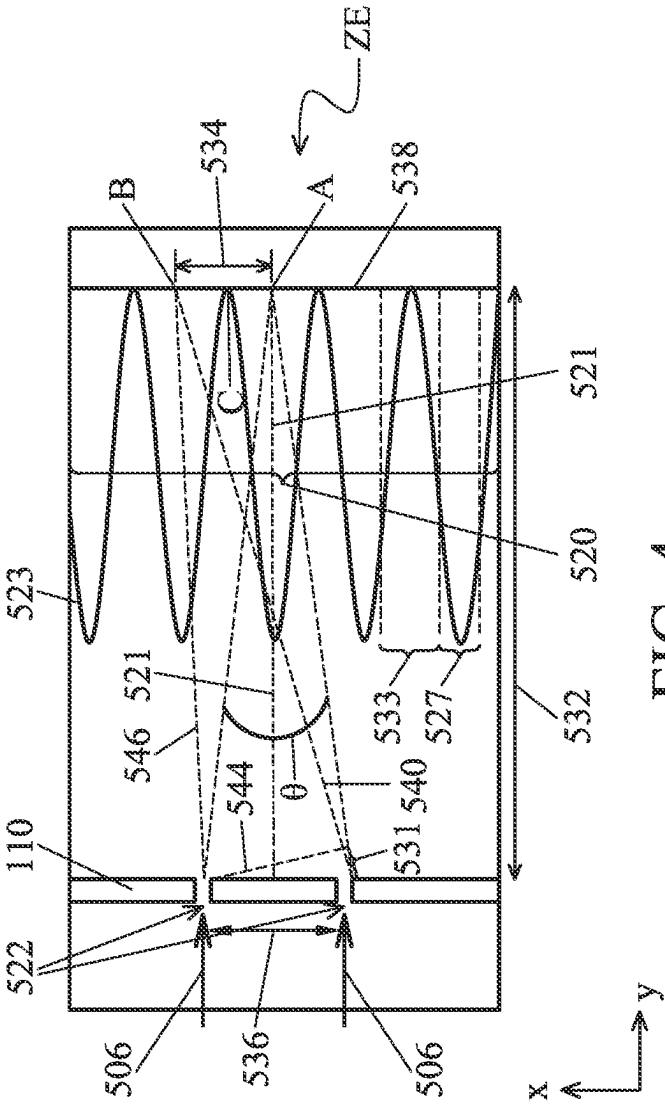


FIG. 4

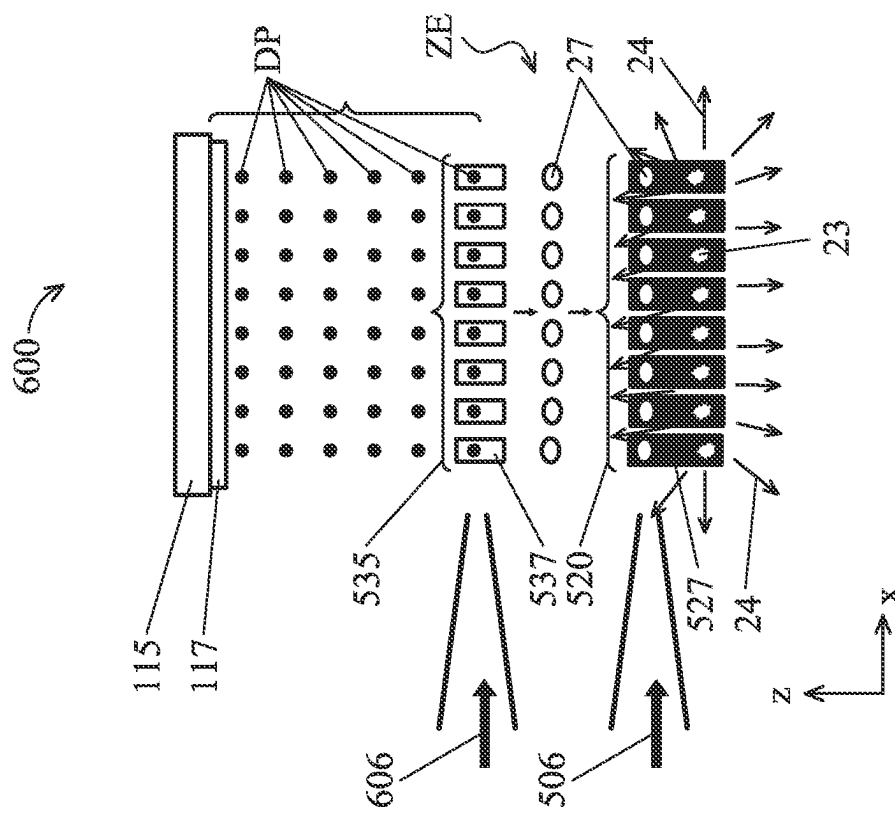


FIG. 6A

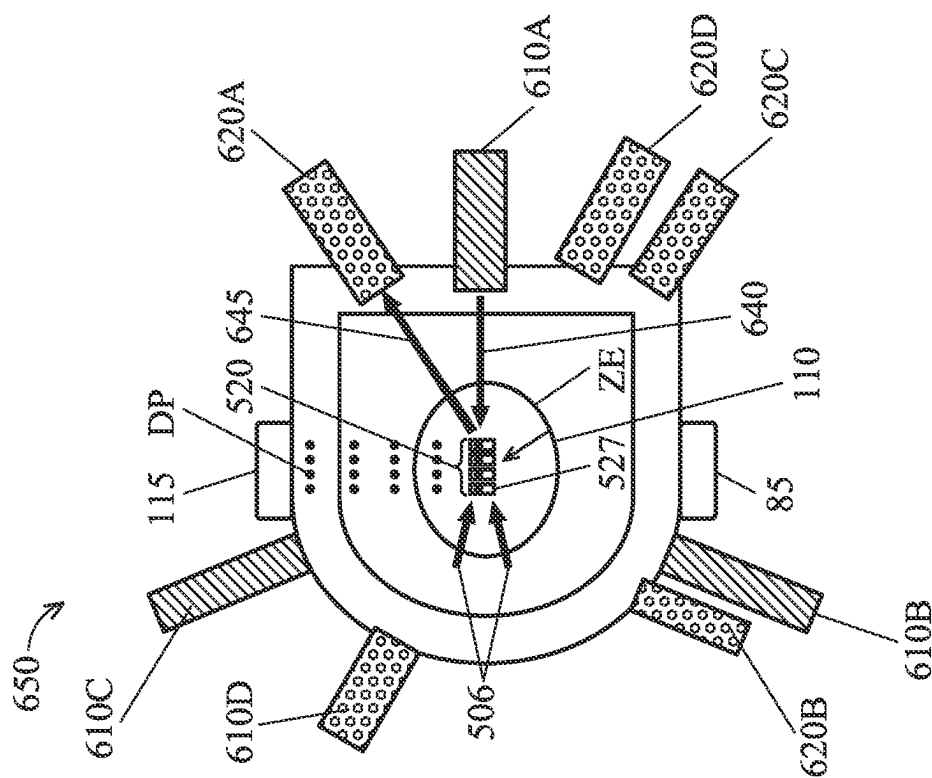


FIG. 6B

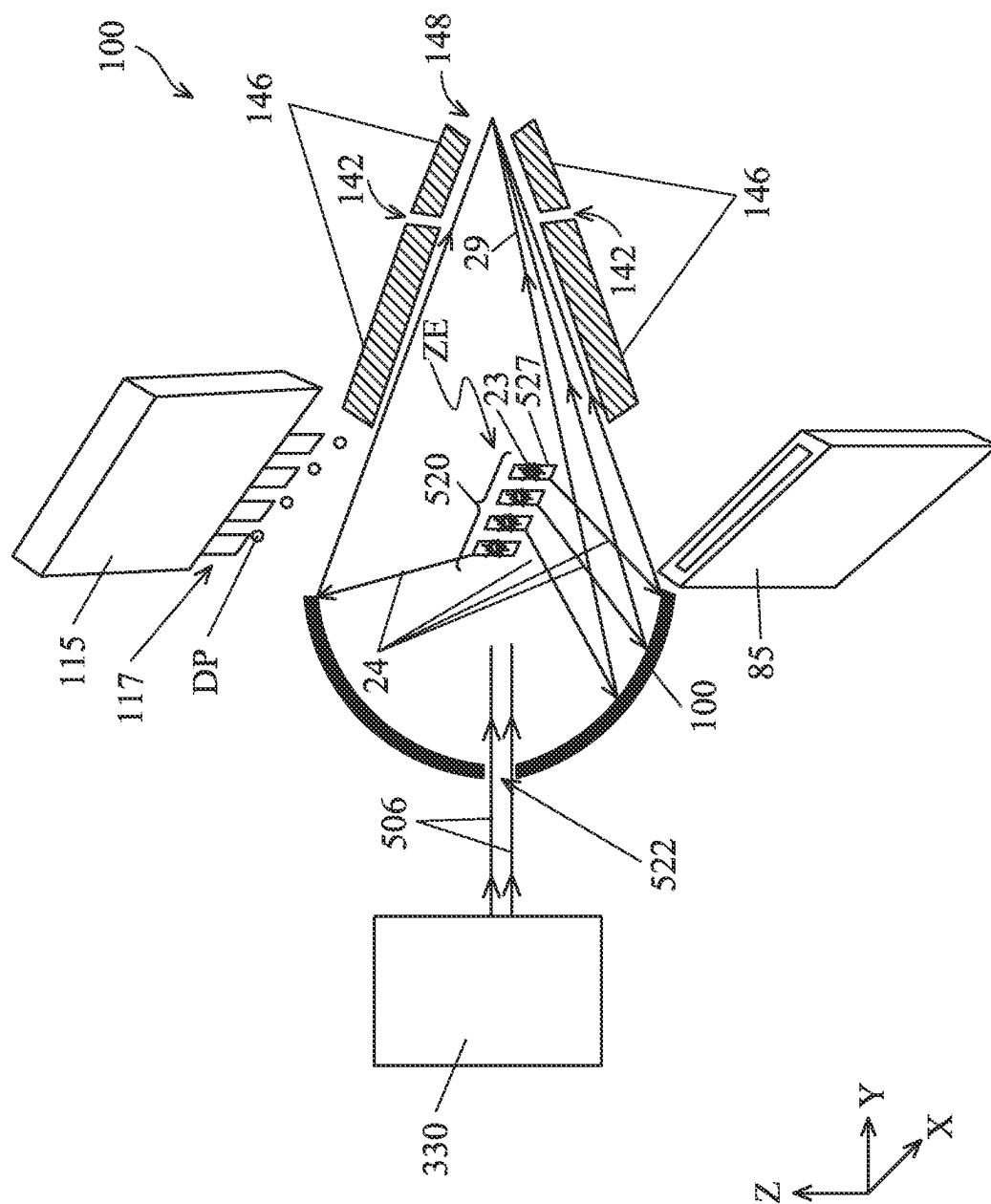


FIG. 7A

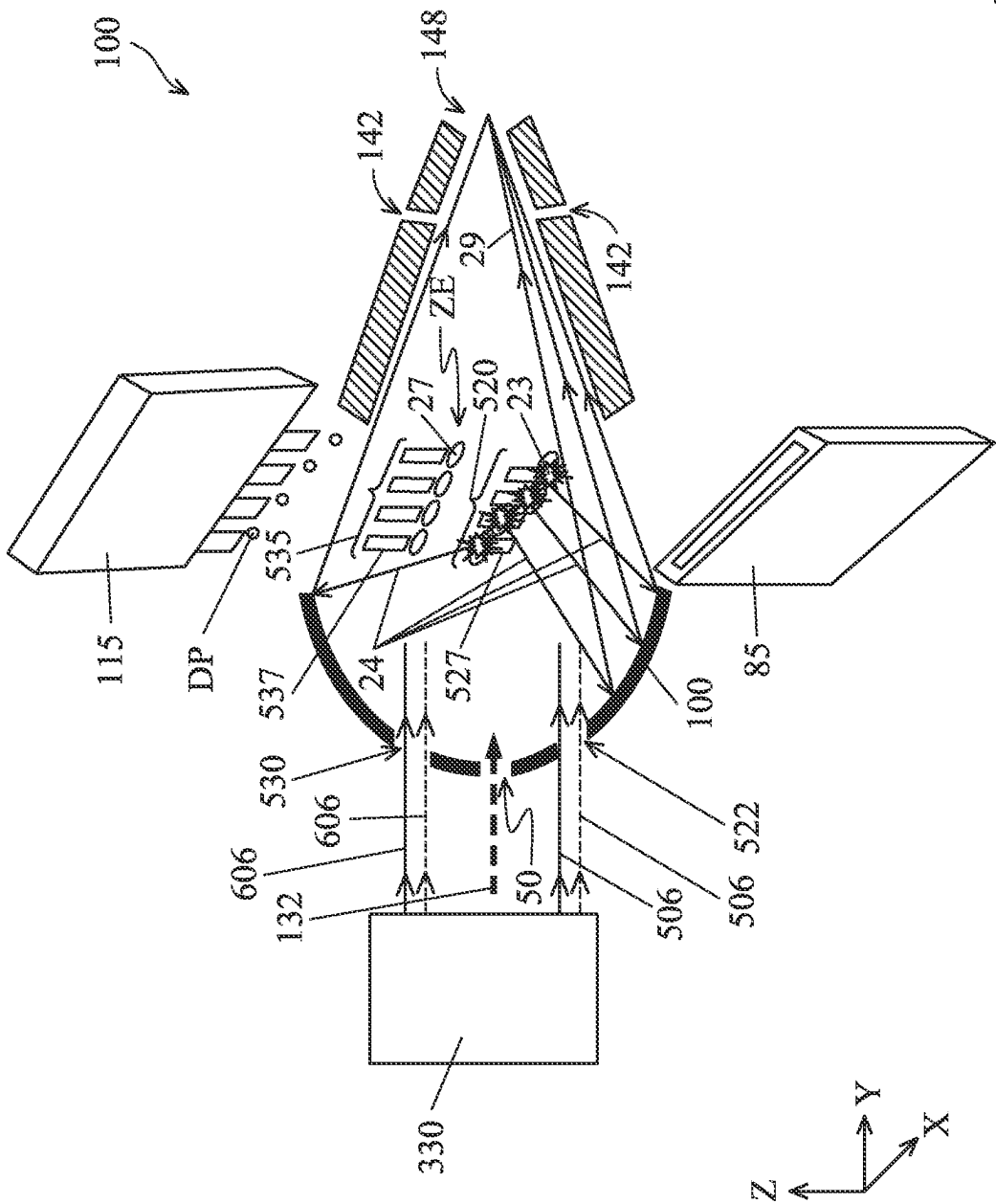


FIG. 7B

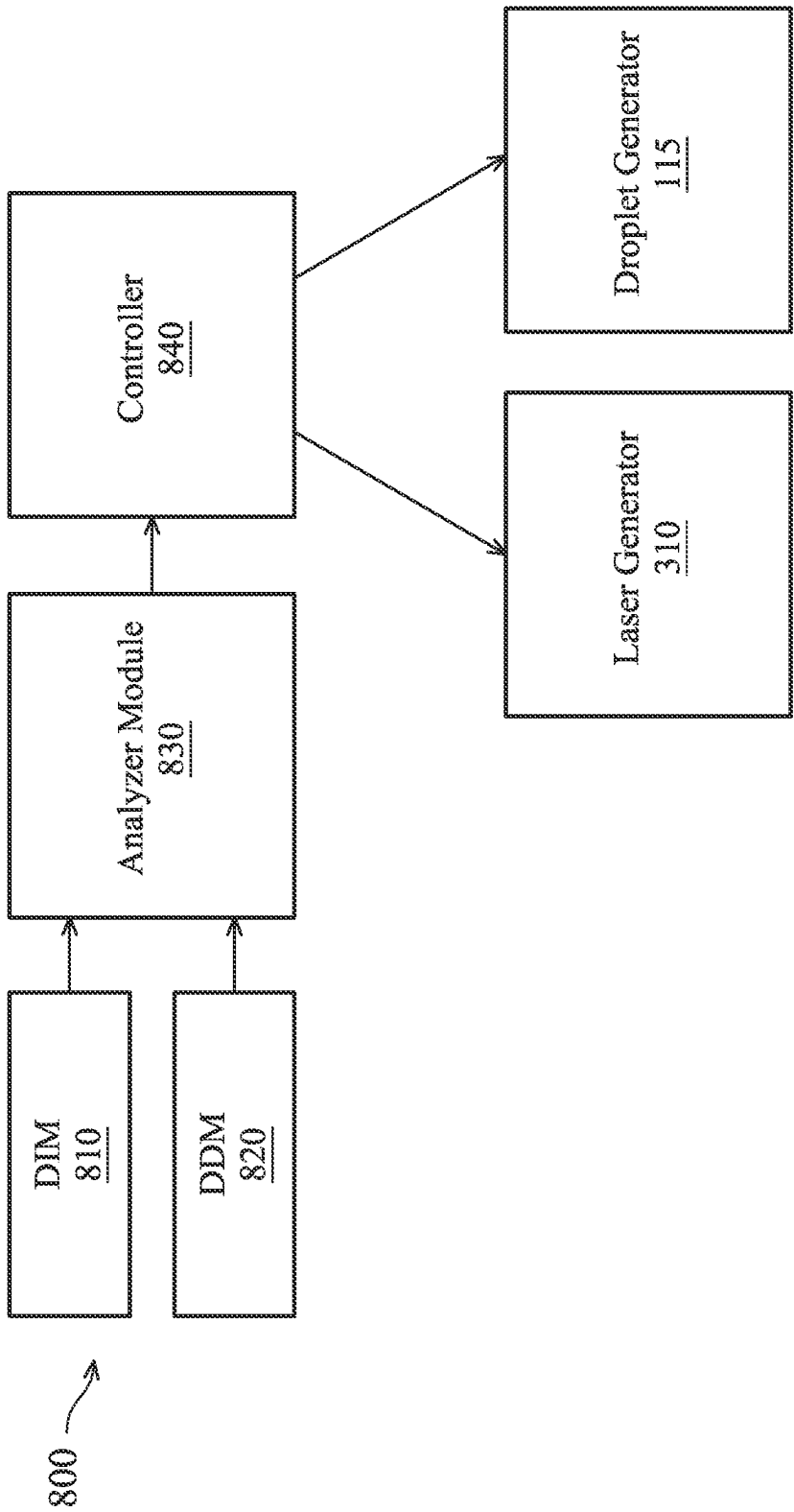


FIG. 8

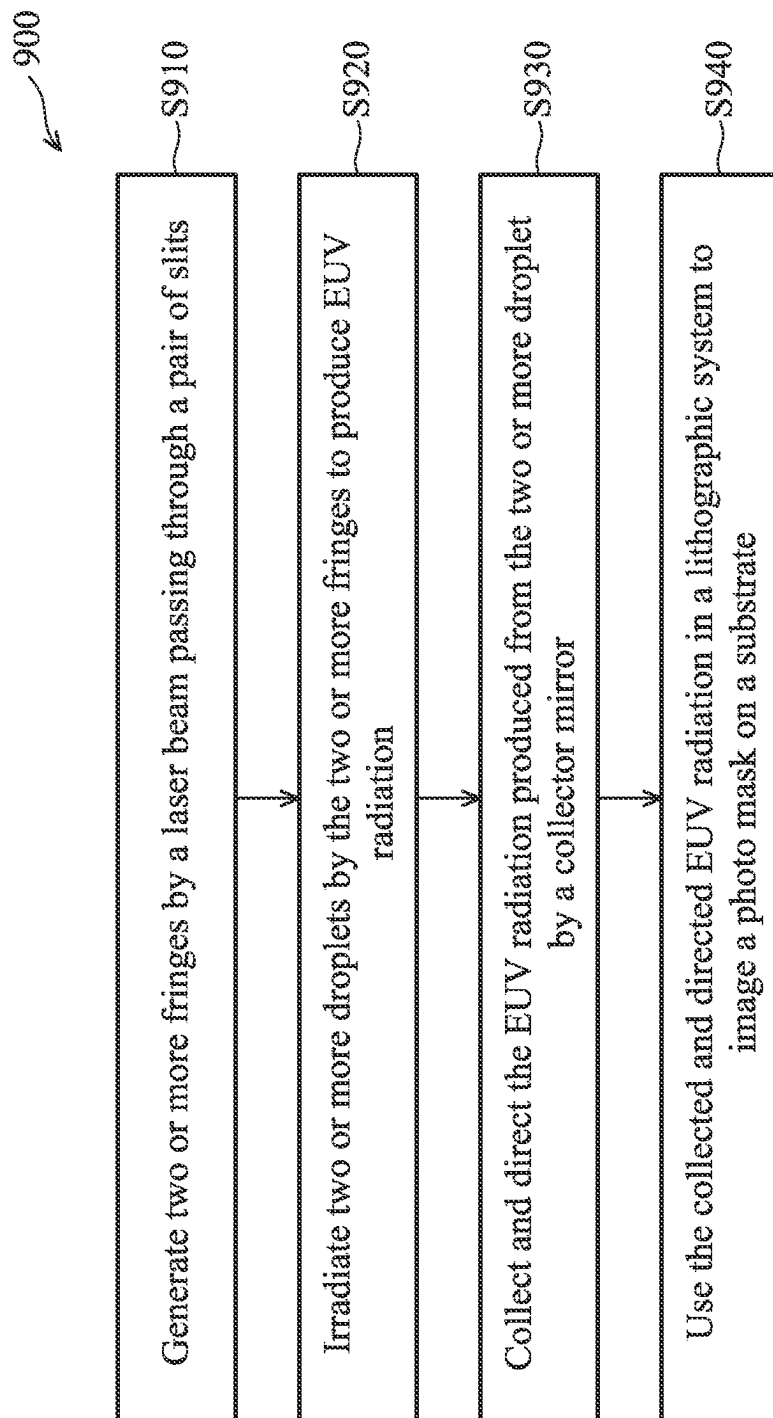


FIG. 9A

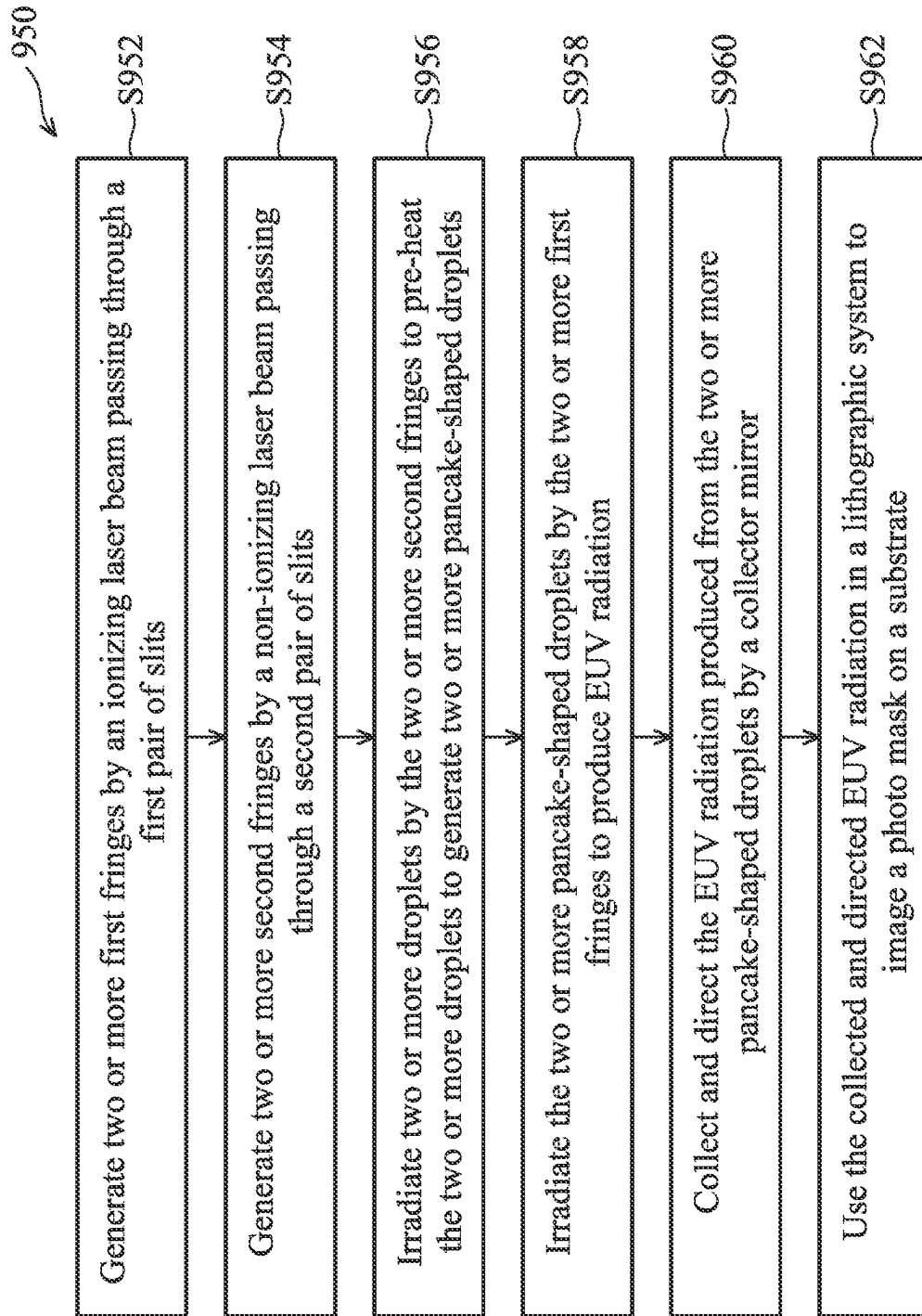


FIG. 9B

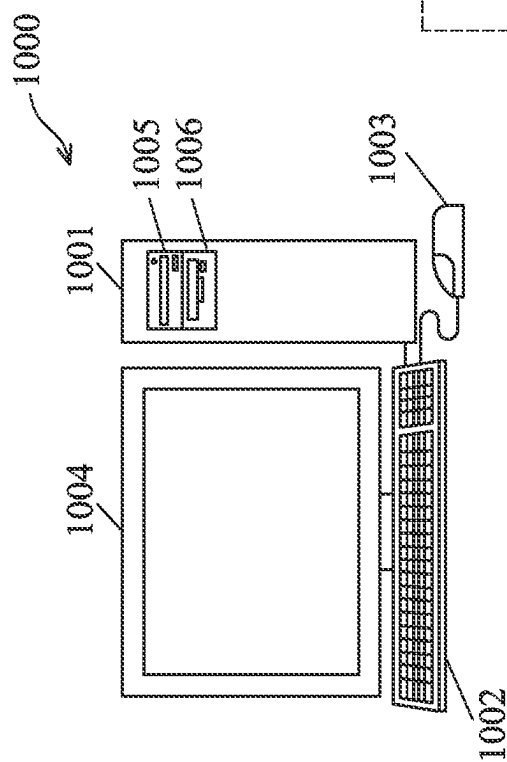


FIG. 10A

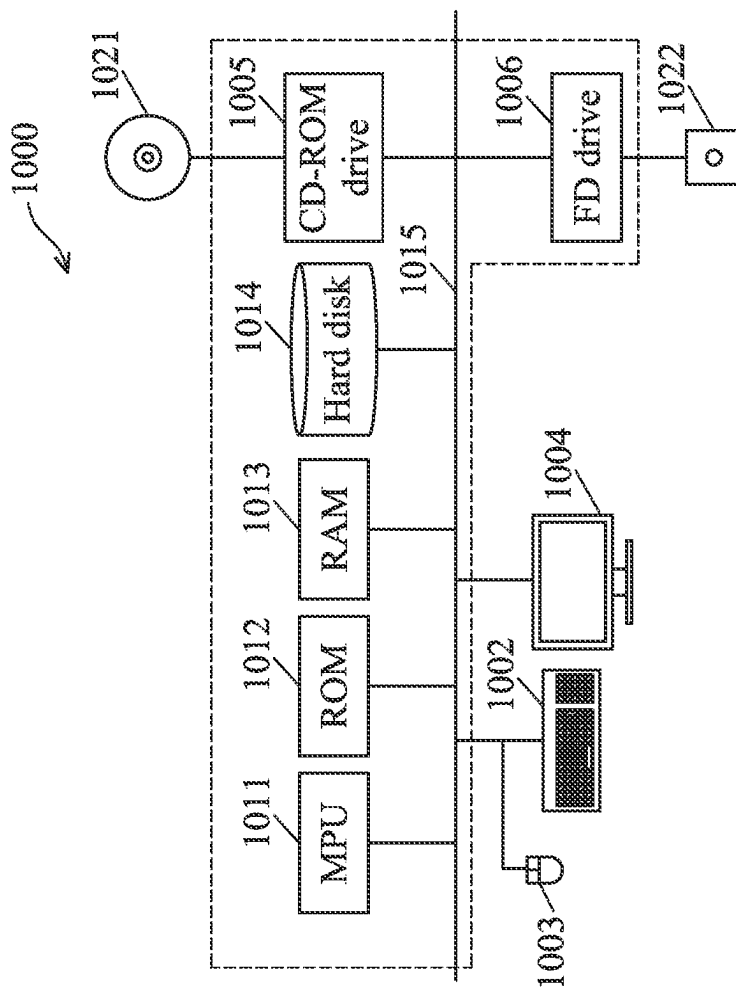


FIG. 10B

METHOD OF MANUFACTURING INTEGRATED CIRCUIT

PRIORITY CLAIM AND CROSS-REFERENCE

This application is a continuation of U.S. patent application Ser. No. 18/114,805 filed Feb. 27, 2023, which is a continuation of U.S. patent application Ser. No. 17/504,210 filed Oct. 18, 2021, now U.S. Pat. No. 11,592,749, which is a continuation of U.S. patent application Ser. No. 16/940,351 filed Jul. 27, 2020, now U.S. Pat. No. 11,150,559, which claims priority to U.S. Provisional Application No. 62/955,245 filed on Dec. 30, 2019, entitled “Laser Interference Fringe Control for Higher EUV Light Source and EUV Throughput,” the entire disclosure of each of which is incorporated herein by reference.

BACKGROUND

The wavelength of radiation used for lithography in semiconductor manufacturing has decreased from ultraviolet to deep ultraviolet (DUV) and, more recently to extreme ultraviolet (EUV). Further decreases in component size require further improvements in resolution of lithography which are achievable using extreme ultraviolet lithography (EUVL). EUVL employs radiation having a wavelength of about 1-100 nm. One method for producing EUV radiation is laser-produced plasma (LPP). In an LPP-based EUV source, a high-power laser beam is focused on small droplet targets of metal, such as tin, to form a highly ionized plasma that emits EUV radiation with a peak maximum emission at 13.5 nm. It is highly desired to increase the efficiency of the LPP-based EUV source.

BRIEF DESCRIPTION OF THE DRAWING

The present disclosure is best understood from the following detailed description when read with the accompanying figures. It is emphasized that, in accordance with the standard practice in the industry, various features are not drawn to scale and are used for illustration purposes only. In fact, the dimensions of the various features may be arbitrarily increased or reduced for clarity of discussion.

FIG. 1 illustrates a schematic view of an EUV lithography system with a laser produced plasma (LPP) EUV radiation source.

FIG. 2 illustrates a schematic view of an EUV lithography exposure tool.

FIGS. 3A and 3B illustrate optical systems of the EUV radiation source for generating a pattern of bright and dark fringes in accordance with some embodiments of the present disclosure.

FIG. 4 illustrates an optical system of the EUV radiation source for generating a pattern of bright and dark fringes in accordance with some embodiments of the present disclosure.

FIG. 5 illustrates a pattern of bright and dark fringes generated by an optical system in accordance with some embodiments of the present disclosure.

FIGS. 6A and 6B illustrate an EUV radiation source for generating plasma from multiple droplets using multiple fringes in accordance with some embodiments of the present disclosure.

FIGS. 7A and 7B illustrate the EUV radiation source for generating EUV radiation from multiple droplets in accordance with some embodiments of the present disclosure.

FIG. 8 schematically illustrates an exemplary EUV lithography system for generating EUV radiation from multiple droplets in accordance with some embodiments of the disclosure.

FIGS. 9A and 9B illustrate flow diagrams of an exemplary processes for generating EUV radiation from multiple droplets in accordance with some embodiments of the disclosure.

FIGS. 10A and 10B illustrate a computer system for an apparatus for generating EUV radiation from multiple droplets in accordance with some embodiments of the disclosure.

DETAILED DESCRIPTION

The following disclosure provides many different embodiments, or examples, for implementing different features of the provided subject matter. Specific examples of components and arrangements are described below to simplify the present disclosure. These are, of course, merely examples and are not intended to be limiting. For example, the formation of a first feature over or on a second feature in the description that follows may include embodiments in which the first and second features are formed in direct contact, and may also include embodiments in which additional features may be formed between the first and second features, such that the first and second features may not be in direct contact. In addition, the present disclosure may repeat reference numerals and/or letters in the various examples. This repetition is for the purpose of simplicity and clarity and does not in itself dictate a relationship between the various embodiments and/or configurations discussed.

Further, spatially relative terms, such as “beneath,” “below,” “lower,” “above,” “upper” and the like, may be used herein for ease of description to describe one element or feature’s relationship to another element(s) or feature(s) as illustrated in the figures. The spatially relative terms are intended to encompass different orientations of the device in use or operation in addition to the orientation depicted in the figures. The apparatus may be otherwise oriented (rotated 90 degrees or at other orientations) and the spatially relative descriptors used herein may likewise be interpreted accordingly. In addition, the term “being made of” may mean either “comprising” or “consisting of” In the present disclosure, a phrase “one of A, B and C” means “A, B and/or C” (A, B, C, A and B, A and C, B and C, or A, B and C), and does not mean one element from A, one element from B and one element from C, unless otherwise described.

An EUV radiation source includes a laser source, a collector mirror, a droplet generator, and a droplet catcher. In some embodiments, the droplet generator includes multiple nozzles and simultaneously generates multiple target droplets of 2, 4, 6, 8, or 10 droplets, in parallel. In some embodiments, the multiple number of droplets are smaller in size and thus require less laser energy to be ionized to produce EUV radiation. In some embodiments, the high power laser beam is divided into a plurality of portions, e.g., divided into two, three, or four portions, and are introduced into the EUV radiation source through a plurality of slits, e.g., openings, and from multiple directions, e.g., through two slits, three slits, or four slits, and from two, three, or four directions. The plurality of portions of the laser beam may interfere at a zone of excitation in the EUV radiation source and may create multiple parallel fringes of laser light at the zone of excitation. In some embodiments, the multiple fringes have a width comparable to a width of the smaller droplets and multiple nozzles of the droplet generator exit the droplets such that each droplet travels along one of the multiple parallel fringes. When the droplet passes through,

e.g. passes along, each fringe, the laser energy of the fringe may ionize the droplets to produce EUV radiation. Although, the laser energy/power is divided into the multiple fringes and thus the energy/power of each fringe is smaller than when the laser beam is focused on one point, the efficiency of producing the EUV radiation may be higher because a surface area of the multiple smaller droplets is more than the surface area of one large droplet having the size of the combination of the smaller droplets. Thus, more laser energy may be transferred to the smaller droplets compared to an amount of laser energy transferred when focusing the laser beam at one point on the large droplet. In addition, the laser fringes have a length such that smaller droplets pass along the fringe length, and thus, stay under the laser beam for a longer time than when the laser beam is focused on one point on the large droplet and. Thus, the efficiency of producing the EUV radiation may further increase.

FIG. 1 shows a schematic view of an EUV lithography system with a laser-produced plasma (LPP) EUV radiation source. The EUV lithography system includes an EUV radiation source 100 (an EUV light source) to generate EUV radiation, an exposure device 200, such as a scanner, and an excitation laser source 300. As shown in FIG. 1, in some embodiments, the EUV radiation source 100 and the exposure device 200 are installed on a main floor MF of a clean room, while the excitation laser source 300 is installed in a base floor BF located under the main floor. Each of the EUV radiation source 100 and the exposure device 200 are placed over pedestal plates PP1 and PP2 via dampers DMP1 and DMP2, respectively. The EUV radiation source 100 and the exposure device 200 are coupled to each other by a coupling mechanism, which may include a focusing unit.

The lithography system is an EUV lithography system designed to expose a resist layer by EUV light (also interchangeably referred to herein as EUV radiation). The resist layer is a material sensitive to the EUV light. The EUV lithography system employs the EUV radiation source 100 to generate EUV light, such as EUV light having a wavelength ranging between about 1 nm and about 100 nm. In one particular example, the EUV radiation source 100 generates an EUV light with a wavelength centered at about 13.5 nm. In the present embodiment, the EUV radiation source 100 utilizes a mechanism of LPP to generate the EUV radiation.

The exposure device 200 includes various reflective optical components, such as convex/concave/flat mirrors, a mask holding mechanism including a mask stage, and wafer holding mechanism. The EUV radiation generated by the EUV radiation source 100 is guided by the reflective optical components onto a mask secured on the mask stage. In some embodiments, the mask stage includes an electrostatic chuck (e-chuck) to secure the mask. Because gas molecules absorb EUV light, the lithography system for the EUV lithography patterning is maintained in a vacuum or a low pressure environment to avoid EUV intensity loss. The exposure device 200 is described in more details with respect to FIG. 2.

In the present disclosure, the terms mask, photomask, and reticle are used interchangeably. In some embodiments, the mask is a reflective mask. In some embodiments, the mask includes a substrate with a suitable material, such as a low thermal expansion material or fused quartz. In various examples, the material includes TiO₂ doped SiO₂, or other suitable materials with low thermal expansion. The mask includes multiple reflective layers (ML) deposited on the substrate. The ML includes a plurality of film pairs, such as

molybdenum-silicon (Mo/Si) film pairs (e.g., a layer of molybdenum above or below a layer of silicon in each film pair). Alternatively, the ML may include molybdenum-beryllium (Mo/Be) film pairs, or other suitable materials that are configurable to highly reflect the EUV light. The mask may further include a capping layer, such as ruthenium (Ru), disposed on the ML for protection. The mask further includes an absorption layer, such as a tantalum boron nitride (TaBN) layer, deposited over the ML. The absorption layer is patterned to define a layer of an integrated circuit (IC). Alternatively, another reflective layer may be deposited over the ML and is patterned to define a layer of an integrated circuit, thereby forming an EUV phase shift mask.

The exposure device 200 includes a projection optics module for imaging the pattern of the mask on to a semiconductor substrate with a resist coated thereon secured on a substrate stage of the exposure device 200. The projection optics module generally includes reflective optics. The EUV radiation (EUV light) directed from the mask, carrying the image of the pattern defined on the mask, is collected by the projection optics module, thereby forming an image on the resist.

In various embodiments of the present disclosure, the semiconductor substrate is a semiconductor wafer, such as a silicon wafer or other type of wafer to be patterned. The semiconductor substrate is coated with a resist layer sensitive to the EUV light in presently disclosed embodiments. Various components including those described above are integrated together and are operable to perform lithography exposing processes. The lithography system may further include other modules or be integrated with (or be coupled with) other modules.

As shown in FIG. 1, the EUV radiation source 100 includes a droplet generator 115 and a LPP collector mirror 110, enclosed by a chamber 105. A droplet DP that does not interact goes to a droplet catcher 85. The droplet generator 115 generates a plurality of target droplets DP, which are supplied into the chamber 105 through a nozzle 117. In some embodiments, the target droplets DP are metal droplets such as tin (Sn), lithium (Li), or an alloy of Sn and Li. Although not shown in the cross-sectional view of FIG. 1, the droplet generator 115 generates multiple target droplets DP simultaneously and in parallel in some embodiments. In some embodiments, as will be described with respect to FIGS. 6A and 6B, the nozzle 117 of the droplet generator 115 has multiple openings that generates multiple target droplets DP in parallel, e.g., parallel target droplets DP. In some embodiments, the target droplets DP each have a diameter in a range from about 10 microns (μm) to about 50 μm. For example, in an embodiment, the target droplets DP are tin droplets, each having a diameter of about 10 μm to about 22 μm. In some embodiments, the target droplets DP are supplied through the multiple openings of the nozzle 117 at a rate in a range from about 50 droplets per second (i.e., an ejection-frequency of about 50 Hz) to about 50,000 droplets per second (i.e., an ejection-frequency of about 50 kHz). For example, in an embodiment, target droplets DP are supplied at an ejection-frequency of about 50 Hz, about 100 Hz, about 500 Hz, about 1 kHz, about 10 kHz, about 25 kHz, about 50 kHz, or any ejection-frequency between these frequencies. The target droplets DP are ejected through the multiple openings of the nozzle 117 and into a zone of excitation ZE (e.g., a target droplet location) at a speed in a range from about 10 meters per second (m/s) to about 100 m/s in various embodiments. For example, in an embodiment, the target droplets DP have a speed of about 10 m/s, about 25 m/s,

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about 50 m/s, about 75 m/s, about 100 m/s, or at any speed between these speeds. Although not shown in the cross-sectional view of FIG. 1, the droplet catcher 85 receives multiple parallel droplets DP that did not interact with the laser beam LR2 in some embodiments. Also, FIGS. 6A and 6B show the droplet catcher 85 that receives the multiple target droplets DP that did not interact. In some embodiments, as will be described with respect to FIGS. 6A and 6B, the nozzle 117 of the droplet generator 115 generates multiple target droplets DP in parallel, e.g., parallel target droplets DP. The parallel target droplets DP may have a diameter in a range from about 10 microns (μm) to about 22 μm . The parallel target droplets DP are ejected through the nozzle 117 and into the zone of excitation ZE at a speed in a range from about 10 meters per second (m/s) to about 100 m/s in various embodiments.

The excitation laser source 300 generates the ionizing laser beam LR2, which is a pulsed beam in some embodiments. The laser pulses of the ionizing laser beam LR2 are generated by the excitation laser source 300. The excitation laser source 300 may include a laser generator 310, laser guide optics 320 and a focusing apparatus 330. In some embodiments, the laser generator 310 includes a carbon dioxide (CO_2) or a neodymium-doped yttrium aluminum garnet (Nd:YAG) laser source with a wavelength in the infrared region of the electromagnetic spectrum. For example, the laser source 310 has a wavelength between 8 μm and 15 μm in an embodiment. The laser beam LR0 generated by the excitation laser source 300 is guided by the laser guide optics 320 and focused, by the focusing apparatus 330, into the ionizing laser beam LR2 that is introduced into the EUV radiation source 100. In some embodiments, in addition to CO_2 and Nd:YAG lasers, the ionizing laser beam LR2 is generated by a gas laser including an excimer gas discharge laser, helium-neon laser, nitrogen laser, transversely excited atmospheric (TEA) laser, argon ion laser, copper vapor laser, KrF laser or ArF laser; or a solid state laser including Nd:glass laser, ytterbium-doped glasses or ceramics laser, or ruby laser. In some embodiments, a non-ionizing laser beam LR1 is also generated by the excitation laser source 300 and the non-ionizing laser beam LR1 is also focused by the focusing apparatus 330. In some embodiments, as will be described with respect to FIGS. 6A and 6B, the ionizing laser beam LR2 is divided into two portions of the laser beam LR2 that reach the zone of excitation ZE from two different angles and thus interact at the zone of excitation ZE to generate a first group of multiple parallel bright fringes at the zone of excitation ZE. At the zone of excitation ZE, each one of the parallel target droplets DP interact, in parallel, with one of the first group of multiple parallel bright fringes to become ionized and to produce EUV radiation.

In some embodiments, the laser source 300 generates the ionizing laser beam LR2 and the non-ionizing laser beam LR1 as pulses. In such embodiments, the non-ionizing laser beam LR1 is a pre-heat laser pulse (interchangeably referred to herein as the "pre-pulse") that is used to heat (or pre-heat) a given target droplet to create a low-density target plume with multiple smaller droplets, which is subsequently heated (or reheated) by a pulse from the ionizing laser beam LR2 (e.g., main pulse), generating increased emission of EUV light compared to when the pre-heat laser pulse is not used.

In some embodiments, as will be described with respect to FIGS. 6A and 6B, the non-ionizing laser beam LR1 is also divided into two portions of laser beam LR1 that reach a location slightly before the zone of excitation ZE from two different angles and thus interact at the location slightly

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before the zone of excitation ZE to generate a second group of multiple parallel bright fringes. In some embodiments, as will be described with respect to FIGS. 6A and 6B, each one of the parallel target droplets DP, interact with one of the second group of multiple parallel bright fringes to get heated and to create the low-density target plume before reaching the zone of excitation ZE. As noted, the laser source 300 generates the laser beams LR1 and LR2 in some embodiments. The laser beam LR1 generates the second group of multiple parallel bright fringes and the laser LR2 generates the first group of multiple parallel bright fringes. In some embodiments, the laser source 300 includes a laser source for generating the laser beam LR2 and also includes another laser source for generating the laser beam LR1. In some embodiments, the laser beam LR0 includes both non-ionizing laser light and ionizing laser light that are generated by the excitation laser source 300 and the laser generator 310. In some embodiments, as described with respect to FIGS. 3B and 7B, the laser guide optics 320 and the focusing apparatus 330 sends the ionizing laser beam LR2 and the non-ionizing laser beam LR1 through separate pairs of slits in the collector mirror 110 to the zone of excitation ZE to generate the first and second group of multiple parallel bright fringes. The second group of multiple parallel bright fringes are generated slightly above the first group of multiple parallel bright fringes.

In various embodiments, the pre-heat non-ionizing laser beam LR1 generates the second group of multiple parallel bright fringes having a width of between about 300 μm to about 500 μm and the ionizing laser beam LR2 generates the first group of multiple parallel bright fringes having a width between about 300 μm to about 500 μm . In some embodiments, the pre-heat non-ionizing laser beam LR1 and the ionizing laser beam LR2 have a pulse-duration in the range from about 10 ns to about 50 ns, and a pulse-frequency in the range from about 1 kHz to about 100 kHz. In various embodiments, the pre-heat laser and the excitation laser have an average power in the range from about 1 kilowatt (kW) to about 50 kW. The pulse-frequency of the non-ionizing laser beam LR1 and the ionizing laser beam LR2 are matched with the ejection-frequency of the target droplets DP in an embodiment.

The ionizing laser beam LR2 is directed through openings (or lenses) into the zone of excitation ZE. The openings adopt a suitable material substantially transparent to the laser beams. The generation of the laser pulses is synchronized with the ejection of the target droplets DP through the nozzle 117. As the target droplets move through the excitation zone, the pre-pulses heat the target droplets and transform them into low-density target plumes. A delay between the pre-pulse of the non-ionizing laser beam LR1 and the main pulse of the ionizing laser beam LR2 is controlled to allow the target plume to form and to expand to an optimal size and geometry. In various embodiments, the pre-pulse and the main pulse have the same pulse-duration and peak power. When the main pulse heats the target plume, a high-temperature plasma is generated. The plasma emits EUV radiation, which is collected by the collector mirror 110. The collector mirror 110, an EUV collector mirror, further reflects and focuses the EUV radiation for the lithography exposing processes performed through the exposure device 200. In some embodiments, as will be described with respect to FIGS. 3A, 3B, 4, and 5, the openings may be one or more pairs of slits such that the two portions of the non-ionizing laser beam LR1 and/or the two portions of the ionizing laser beam LR2 enter from a pair of slits. As described, the interaction of the two portions of the

ionizing laser beam **LR2** generates the first group of multiple parallel bright fringes at the zone of excitation **ZE** and the interaction of the two portions of the non-ionizing laser beam **LR1** generates the second group of multiple parallel bright fringes slightly before the zone of excitation **ZE** in some embodiments.

One method of synchronizing the generation of a pulse (either or both of the pre-pulse and the main pulse) from the excitation laser with the arrival of the target droplet in the zone of excitation **ZE** is to detect the passage of a target droplet at given position and use it as a signal for triggering an excitation pulse (or pre-pulse). In this method, if, for example, the time of passage of the target droplet is denoted by t_o , the time at which EUV radiation is generated (and detected) is denoted by t_{rad} , and the distance between the position at which the passage of the target droplet is detected and a center of the zone of excitation is d , the speed of the target droplet, v_{dp} , is calculated as

$$v_{dp} = d / (t_{rad} - t_o) \quad \text{Equation (1)}$$

Because the droplet generator **115** is expected to reproducibly supply droplets at a fixed speed, once v_{dp} is calculated, the excitation pulse is triggered with a time delay of d/v_{dp} after a target droplet is detected to have passed the given position to ensure that the excitation pulse arrives at the same time as the target droplet reaches the center of the zone of excitation. In some embodiments, because the passage of the target droplet is used to trigger the pre-pulse, the main pulse is triggered following a fixed delay after the pre-pulse. In some embodiments, the value of target droplet speed v_{dp} is periodically recalculated by periodically measuring t_{rad} , if needed, and the generation of pulses with the arrival of the target droplets is resynchronized. The timing and the duration of the pre-pulse and/or the main pulse when the first and/or the second group of multiple parallel bright fringe are generated are described below with respect to FIGS. **6A** and **6B**.

As shown in FIG. **1**, a buffer gas is supplied from a first buffer gas supply **130** through the opening in collector mirror **110** by which the ionizing laser beam **LR2** and/or the non-ionizing laser beam **LR1** is delivered to the tin droplets. In some embodiments, the buffer gas is H_2 , He, Ar, N or another inert gas. In certain embodiments, H_2 is used as the buffer gas, as H radicals that are generated by ionization of the buffer gas can be used for cleaning purposes. The buffer gas can also be provided through one or more second buffer gas supplies **135** toward the collector mirror **110** and/or around the edges of the collector mirror **110**. Further, the chamber **105** includes one or more gas outlets **140** so that the buffer gas is exhausted outside the chamber **105**. Hydrogen gas has low absorption to the EUV radiation. Hydrogen gas reaching to the coating surface of the collector mirror **110** reacts chemically with a metal of the droplet forming a hydride, e.g., metal hydride. When tin (Sn) is used as the droplet, stannane (SnH_4), which is a gaseous byproduct of the EUV generation process, is formed. The gaseous SnH_4 is then pumped out through the gas outlet **140**. However, it is difficult to exhaust all gaseous SnH_4 from the chamber and to prevent the SnH_4 from entering the exposure device **200**. Therefore, monitoring and/or control of the debris in the EUV radiation source **100** is beneficial to the performance of the EUVL system.

FIG. **2** shows a schematic view of an EUV lithography exposure tool. The EUVL exposure tool of FIG. **2** includes

the exposure device **200** that shows the exposure of photoresist coated substrate, a target semiconductor substrate **210**, with a patterned beam of EUV light. The exposure device **200** is an integrated circuit lithography tool such as a stepper, scanner, step and scan system, direct write system, device using a contact and/or proximity mask, etc., provided with one or more optics **205a**, **205b**, for example, to illuminate a patterning optic, such as a reticle, e.g., a reflective mask **205c**, with a beam of EUV light, to produce a patterned beam, and one or more reduction projection optics **205d**, **205e**, for projecting the patterned beam onto the target semiconductor substrate **210**. A mechanical assembly (not shown) may be provided for generating a controlled relative movement between the target semiconductor substrate **210** and patterning optic, e.g., the reflective mask **205c**. As further shown, the EUVL exposure tool of FIG. **2**, further includes the EUV radiation source **100** including a plasma plume **23** at the zone of excitation **ZE** emitting EUV light in the chamber **105** that is collected and reflected by a collector mirror **110** into the exposure device **200** to irradiate the target semiconductor substrate **210**. In some embodiments, the first and second group of multiple parallel bright fringes are in a small area of about $500 \mu m$ by about $500 \mu m$ in the zone of excitation **ZE** of FIG. **2** and the collector mirror **110** is adjusted such that the EUV radiation generated in the EUV radiation source **100** is gathered and focused by the collector mirror **110** and is transmitted outside the EUV radiation source **100** to the one or more optics **205a**, **205b**.

FIGS. **3A** and **3B** illustrate optical systems of the EUV radiation source for generating a pattern of bright and dark fringes in accordance with some embodiments of the present disclosure. In some embodiments, the optical system **350** of FIG. **3A** is implemented in the EUV radiation source **100** of FIG. **1** and uses the laser beam **LR0** generated by the excitation laser source **300** to generate the pattern of bright and dark fringes in the zone of excitation **ZE** of the EUV radiation source **100**.

FIG. **3A** shows a plan view in the XY-plane of an optical system **350** that receives a laser beam **512** that is consistent with the laser beam **LR0** of FIG. **1**. The laser beam **512** enters a mirror system **504** that includes a beam splitter **510** and three mirrors **508**, e.g., 45 degree mirrors. The mirror system **504** divides the laser beam **512** into two laser beams **506**, a pair of laser beams **506**, having similar, e.g., essentially equal, intensity. The pair of laser beams **506** pass through a pair of slits **522** in the collector mirror **110** and generate a pattern of bright and dark fringes **520** having fringes **527** in the excitation zone **ZE** in front of the slits **522** in the XZ-plane. As shown in FIG. **3A**, the pair of laser beams **506** do not travel the same distance to reach the zone of excitation **ZE** and, thus, a bright fringe **527** may not occur at a midline **521**, e.g., a symmetry line, between the pair of slits **522**. In some embodiments, the laser beam **512** is consistent with the laser beam **LR2** or laser beam **LR1** of FIG. **1**. In some embodiments, the laser beam **512** is consistent with the laser beam **506** or laser beam **606** of FIGS. **6A**, **6B**, **7A**, and **7B**.

In some embodiments, as shown in FIG. **3A**, a delay segment **502** is introduced in the path of one of the laser beams **506** to generate a delay in one of the laser beams **506** and, thus, move the bright and dark fringes. In some embodiments, the delay segment **502** generates a delay such that the laser beams **506** are in-phase and, thus, a bright fringe **527** is generated at the midline **521** in the zone of excitation **ZE**. In some embodiments, the delay segment **502** generates a delay such that the laser beams **506** are 180 degrees out of phase and, thus, a dark fringe **533** is generated

at the midline **521** in the zone of excitation ZE. In some embodiments, the pair of slits **522** are rectangular openings in the collector mirror **110**. In some embodiments, a distance **536**, in the X-direction, between the center of slits **522** is between about 0.5 mm to about 10 mm. The slits are described in more details with respect to FIG. 3B. As shown in FIG. 3A, the collector mirror **110** is a cross-section of the collector mirror and, thus, the pattern of bright and dark fringes **520** extend in the Z-direction that is perpendicular to the XY-plane of FIG. 3A.

FIG. 3B shows two pair of slits **522** and **530** in the collector mirror **110**. FIG. 3B is a plan view of the collector mirror **110** that is perpendicular to the Y-direction. As shown, the pair of slits **530** is arranged on top of the pair of slits **522** in the Z-directions. As described, the collector mirror **110** may have an elliptical or a parabolic shape to direct the generated EUV radiation from the EUV radiation source **100**. In some embodiments, each slit of the pairs of slits has a rectangular shape with a width **554** and a length **552** and the distance **536** separating the centers of the slits in the X-direction. In some embodiments, the slits of each pair of slits have similar shape and size, while the slits of the pair of slits **522** and slits of the pair of slits **530** have a different shape and/or size. In some embodiments, the laser beam LR2 is divided by the mirror system **504** into a pair of laser beams that passes through the pair of slits **522** and creates a pattern of bright and dark fringes in the zone of excitation ZE. In some embodiments, the laser beam LR1 is divided by another mirror system (not shown) into a pair of laser beams that passes through the pair of slits **530** and creates a pattern of bright and dark fringes in the zone of excitation ZE. In some embodiments, the laser beam LR2 creates a first pattern of bright and dark fringes in the zone of excitation ZE and the laser beam LR1 creates a second pattern of bright and dark fringes in the zone of excitation ZE, which is above the first pattern of bright and dark fringes. Thus, two non-overlapping patterns of bright and dark fringes may be created one on top of the other by the laser beams LR1 and LR2 at the zone of excitation ZE. In some embodiments, the distance **536** is between about 0.5 mm and about 10 mm. The width **554** is between about 200 nm and about 10 μ m, and the length **552** is between about 100 μ m and about 1 mm. In some embodiments, the length **552** and the width **554** of the pair of slits **522** and **530** are the same, however, the distance **536** between the two pair of slits **522** and **530** are not the same because the wavelength of the laser beams LR1 and LR2 are not the same. In some embodiments, the distance **536** between the slits of the pairs of slits **530** is adjusted by the ratio of the wavelength of the laser beams LR2 to LR1 such that both of the first and second group of multiple parallel bright fringes generated by the laser beams LR2 to LR1 have the same effective width of the fringes and the same effective distance between adjacent fringes.

FIG. 4 illustrates an optical system of the EUV radiation source for generating a pattern of bright and dark fringes in accordance with some embodiments of the present disclosure. FIG. 4 shows an optical system **400** that receives the laser beams **506** of FIG. 3A. The laser beams **506** enter the slits **522** in the collector mirror **110** and generate the pattern of bright and dark fringes **520** over a line **538** at the zone of excitation ZE along the X-direction. An intensity variation of the pattern of bright and dark fringes **520** is shown by curve **523**. In some embodiments, as shown in FIG. 4, the curve **523** has maxima and minima. The curve **523** has a maximum at point A where the midline **521** intersects the line **538** at the zone of excitation ZE. In some embodiments,

the laser beams **506** entering the pair of slits **522** are in-phase and, thus, the beams of light reaching point A from the pair of slits **522** remain in-phase and constructively add at point A to generate a maximum at point A. In some embodiments, the distance **536** between the slits is between 200 nm and 100 μ m. In some embodiments, the pair of slits **522** are next to each other and an angle θ between the incident beams is between 0.5 degrees and 5 degrees. In some embodiments, the slits **522** are at opposite sides of the collector mirror **110** and the angle θ between the incident beams is between 175 degrees and 179.5 degrees.

Similarly, the laser beams **506** entering the pair of slits **522** travel the paths **546** and **540** to reach a point B on the line **538** at the zone of excitation ZE. In some embodiments, the paths **546** and **540** have a travel path difference **531**, which is shown by a perpendicular line **544** from the upper slit **522** to the path **540** in FIG. 4. The travel path difference **531** is equal to one or more wavelengths, e.g., one wavelength, of the laser beams **506**. Thus, the beams of light reaching point B from the pair of slits **522** remain in-phase and constructively add at point B to generate a maximum at point B. In some embodiments, a length **534** between point A and point B is equal to the wavelength of the laser beams **506** multiplied by a ratio of a distance **532** over the distance **536**. The distance **532** is a distance between the collector mirror **110** and the zone of excitation ZE in the Y-direction and the distance **536** is the distance between the slits **522** of the collector mirror **110** in the X-direction. Conversely, the laser beams **506** entering the pair of slits **522**, destructively add at point C between the points A and B and, thus, a minimum is generated at point C. In some embodiments, as shown in FIG. 4, the curve **523** has cosine squared shape and, as shown, the bright fringe **527** is centered around a maximum point of the curve **523** and the dark fringe **533** is centered around a minimum point of the curve **523**.

FIG. 5 illustrates a pattern of bright and dark fringes generated by an optical system in accordance with some embodiments of the present disclosure. FIG. 5 shows the pattern of bright and dark fringes **520** at the zone of excitation ZE and also shows the bright fringes **527** and the dark fringes **533**. In some embodiments, the pattern of bright and dark fringes **520** extends along the line **538** in the X-direction in a width **519**. In some embodiments, each fringe of the pattern of bright and dark fringes **520** extends along a line perpendicular to the line **538**, e.g., in the Z-direction, in a length **525**. As shown in FIG. 4, the length **534** is a sum of the widths of a bright fringe **527** and a dark fringe **533** and, thus, the width of each one of the bright fringes **527** and the width of each one of the dark fringes **533** is half of the length **534**. In some embodiments, the length of the bright fringes **527** shrinks as the bright fringes **527** are further away from the center of the pattern of bright and dark fringes **520**. In some embodiments, when the slits **522** have a rectangular shape in the XZ-plane, the pattern of bright and dark fringes **520** has a 2D sinc-function envelope in the XZ-plane at the zone of excitation ZE. Thus, the pattern of bright and dark fringes **520** is a 2D cosine squared function that changes in the X-direction and is limited by a main lobe of the 2D sinc-function in some embodiments. In some embodiments, the length **525** of the fringes decrease as the fringe gets farther from the central fringe. Thus, the fringes farther from the central fringe do not have the effective length such that when the droplet DP passes through the fringe enough energy of the laser beams LR1 or LR2 is transferred to the droplet DP. Thus, the length **525** of the fringes defines the number of parallel droplets that simultaneously pass through the fringes.

In some embodiments, the width **519** of the pattern of bright and dark fringes **520** is inversely proportional to the width **554** of the pair of slits **522** and the length **525** of the pattern of bright and dark fringes **520** is inversely proportional to the length **552** of the pair of slits **522**. In some embodiments, the 2D sinc-function has other less intense lobes in the X-direction and the Z-direction and thus other patterns of bright and dark fringes repeat, with less intensity, right, left, above, and below the pattern of bright and dark fringes **520** and at least a portion of the other patterns of bright and dark fringes is not visible. In some embodiments, a frequency of the fringes of the pattern of bright and dark fringes **520**, e.g., the frequency of the cosine squared function, is proportional to the distance **536** between the center of the pair of slits **522**. In some embodiments, the fringes in a lobe between the droplet generator **115** and the pattern of bright and dark fringes **520** is used for pre-heating the droplets DP.

FIGS. **6A** and **6B** illustrate an EUV radiation source for generating plasma from multiple droplets using multiple fringes in accordance with some embodiments of the present disclosure. FIGS. **6A** and **6B** show the droplet generator **115** having the nozzle **117** that simultaneously produces multiple droplets DP in parallel. FIG. **6A** shows the non-ionizing laser beam LR1 that passes through a mirror system consistent with the mirror system **504** of FIG. **3A** generates the pair of laser beams **606**. In addition, the pair of laser beams **606** passes through a pair of slits consistent with the pair of slits **530** of FIG. **3B** and generates a pattern of bright and dark fringes **535** having fringes **537** at the zone of excitation ZE. When reaching the excitation zone ZE, the multiple parallel droplets DP pass through the pattern of bright and dark fringes **535** and thus a pancake-shaped droplet **27** is generated from each one of the multiple droplets DP. As shown in FIG. **6A**, each one of the multiple droplets DP pass through one of the bright fringes **537** of the pattern of bright and dark fringes **535**.

FIG. **6A** also shows the ionizing laser beam LR2 that passes through a mirror system consistent with the mirror system **504** of FIG. **3A** generates the pair of laser beams **506**. In addition, the pair of laser beams **506** passes through a pair of slits consistent with the pair of slits **522** of FIG. **3B** and generates the pattern of bright and dark fringes **520** at the zone of excitation ZE. The pancake-shaped droplets **27** pass through the pattern of bright and dark fringes **520** such that each pancake-shaped droplet **27** passes through one of the bright fringes **527** of the pattern of bright and dark fringes **520**. Passing through the bright fringes **527**, each pancake-shaped droplet **27** interacts with the bright fringes **527** at the zone of excitation ZE in the chamber of the EUVL system to form the plasma plume **23** which emits EUV light rays **24** in all directions. In some embodiments, the multiple droplets DP simultaneously interact at the excitation zone ZE by the multiple bright fringes **527** that are generated by the ionizing laser beam LR2 to produce the plasma plume **23** which emits EUV light rays **24** in all directions.

FIG. **6B** shows the pair of laser beams **506** passes through a pair of slits consistent with the pair of slits **522** of FIG. **3B** in the collector mirror **110** and generates the pattern of bright and dark fringes **520** at the zone of excitation ZE. The multiple droplets DP pass through the pattern of bright and dark fringes **520** such that each droplet DP passes through one of the bright fringes **527** of the pattern of bright and dark fringes **520**. Passing through the bright fringes **527**, droplets DP interact with the bright fringes **527** at the zone of

excitation ZE in the chamber of the EUVL system to form the plasma plume **23** which emits EUV light rays **24** in all directions.

As shown in FIGS. **6A** and **6B**, each one of the multiple droplets DP receives the laser energy when passing through the bright fringes and thus receives laser energy for a longer amount of time compared to the amount of time the droplet DP receives the laser energy in the EUV radiation source **100** of FIG. **1**. Thus, although the ionizing laser beam LR2 is divided into fringes in FIGS. **6A** and **6B**, by irradiating multiple droplets DP simultaneously and receiving the laser energy for a longer time, the efficiency of the EUV radiation source of FIGS. **6A** and **6B** is greater than the efficiency of the EUV radiation source **100** of FIG. **1** for transferring the laser energy into EUV radiation.

In some embodiments, as shown in an EUV source **650** of FIG. **6B** one or more droplet illumination modules (DIM) **610A**, **610B**, **610C**, or **610D** irradiates the zone of excitation ZE with non-ionizing light to illuminate the droplets DP. In addition, one or more droplet detection module (DDM) **620A**, **620B**, **620C**, or **620D** take images of the zone of excitation ZE to determine, e.g., calculate, the velocity of the multiple droplets DP and to determine the timing of the laser pulses that generate the ionizing laser beam LR2 and the timing of the laser pulses that generate the non-ionizing laser beam LR1 such that the pattern of bright and dark fringes **520/535** is generated when the multiple droplets DP/pancake-shaped droplets **27** are respectively traveling through the pattern of bright and dark fringes **520/535**. In some embodiments, the DIM **610A** transmits an illumination radiation beam **640** to the zone of excitation ZE and the DDM **620A** receives a reflectance beam **645** from the zone of excitation ZE.

In some embodiments, the collector mirror **110** is modified, e.g., adjusted, such that the interactions that generates the plasma plumes **23**, which emits EUV light rays **24** from each one of the multiple droplets DP, in the zone of excitation ZE and the EUV light rays **24** are gathered and directed by the collector mirror **110** to the exposure device **200**. In some embodiments, the pattern of bright and dark fringes **520** occurs in the zone of excitation ZE and the zone of excitation ZE is at or sufficiently around a focal point of the collector mirror **110**. In some embodiments, the collector mirror **110** is adjusted to improve the efficiency of the collector mirror **110** for gathering and focusing the generated EUV radiation. In some embodiments, a size of the multiple droplets DP in FIGS. **6A** and **6B** is reduced compared to the size of the droplet DP in FIG. **1**, and thus, the droplets of FIGS. **6A** and **6B** produce the plasma plume **23** using less laser energy compared to producing the plasma plume **23** in FIG. **1**. In some embodiments, as noted, the size of the multiple droplet DP is between about 10 μm to about 22 μm . In some embodiments, the size and the number of the multiple parallel droplets DP are adjusted such that EUV radiation of 250 Watts is generated.

FIGS. **7A** and **7B** illustrate the EUV radiation source for generating EUV radiation from multiple droplets in accordance with some embodiments of the present disclosure. FIG. **7A** shows the details of the droplet generator **115** of FIG. **1** that has the nozzle **117** that simultaneously produces multiple droplets DP from multiple adjacent openings, e.g., in parallel. The ionizing laser beam LR2 is divided into the pair of laser beams **506** by a mirror system consistent with the mirror system **504** of FIG. **3A**. The pair of laser beams **506** pass through the pair of slits **522** of the collector mirror **110** and produce the pattern of bright and dark fringes **520** in an XZ-plane in front of the pair of slits **522** in the zone

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of excitation ZE. The multiple droplets DP pass through the plane of the pattern of bright and dark fringes 520 such that each DP passes through one bright fringe 527 of the pattern of bright and dark fringes 520 receiving the laser beam energy from the bright fringe and producing the plasma plumes 23 that generate the EUV light rays 24 (EUV radiation). The droplet catcher 85 is configured to receive the debris droplets associated with the multiple droplets DP. In some embodiments, the collector mirror 110 collects and directs the generated EUV light rays 24 to the opening 148 of the EUV radiation source 100. In some embodiments, the ionizing laser beam LR2 interacts with the tin droplets DP at the zone of excitation ZE in a space between collector mirror 110 and the walls 146 of EUV radiation source 100 to form the plasma plume 23 which emits EUV light rays 24 in all directions. The walls 146 are used to create a cone shape such that the EUV radiation along arrows 29 inside the cone shape exit through the opening 148 and any other radiation that is not along the arrows 29 do not exit the cone shape and thus do not exit the EUV radiation source 100.

FIG. 7B also shows the droplet generator 115 that has the nozzle 117 that simultaneously produces multiple droplets DP from multiple adjacent openings in parallel. The ionizing laser beam LR2 is divided into the pair of laser beams 506 by a mirror system consistent with the mirror system 504 of FIG. 3A. The pair of laser beams 506 pass through the pair of slits 522 of the collector mirror 110 and produce the pattern of bright and dark fringes 520 in an XZ-plane in front of the pair of slits 522 in the zone of excitation ZE. The pre-heat non-ionizing laser beam LR1 is divided into the pair of laser beams 606 by a mirror system consistent with the mirror system 504 of FIG. 3A. The pair of laser beams 606 pass through the pair of slits 530 of the collector mirror 110 and produce the pattern of bright and dark fringes 535 in an XZ-plane in front of the pair of slits 530. In some embodiments, the pattern of bright and dark fringes 535 and the pattern of bright and dark fringes 520 are non-overlapping and occur in the same XZ-plane in the zone of excitation ZE. In some embodiments, an opening 50 of the collector mirror 110 is used for a gas flow 132 to enter into the EUV radiation source 100. In some embodiments, at least a portion of the gas flow 132 exits through openings 142 in the wall 146 of the cone shape.

As described with respect to FIG. 6A, the multiple droplets DP pass through the plane of the pattern of bright and dark fringes 535 and 520 such that each DP first passes through one bright fringe 537 of the pattern of bright and dark fringes 535, receives the laser beam energy from the bright fringe 537, and produces the pancake-shaped droplets 27. Then the pancake-shaped droplets 27 each passes through one bright fringe 527 of the pattern of bright and dark fringes 520, receives the laser beam energy from the bright fringe 527, and produces the plasma plumes 23 that produce the EUV light rays 24 (EUV radiation). Thus, the multiple droplets DP first pass through the bright fringes 537 of the pattern of bright and dark fringes 535, receive the pre-heat non-ionizing energy of the laser beam LR1, and produce the pancake-shaped droplets 27. Then, the pancake-shaped droplets 27 pass through the bright fringes 527 of the pattern of bright and dark fringes 520, receive the ionizing energy of the laser beam LR2 and produce the plasma plumes 23 that generate the EUV light rays 24. In some embodiments, the collector mirror 110 collects and directs the generated EUV light rays 24 to the opening 148 of the EUV radiation source 100. In some embodiments, the size of the droplets DP, the number of parallel droplets DP, and the distance between the parallel droplets are designed such that

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during each pulse of the laser beams LR1 and LR2, the multiple droplets DP simultaneously pass through the first group of multiple parallel bright fringes 537 to produce the pancake-shaped droplets 27. The pancake-shaped droplets 27 simultaneously pass through the second group of multiple parallel bright fringes 527 to generate EUV radiation while the droplets DP and the pancake-shaped droplets 27 are inside the fringes. In some embodiments, the effective width of a fringe is half of the distance 536 and the effective distance between fringes is also half of the distance 536.

FIG. 8 schematically illustrates an exemplary EUV lithography system for generating EUV radiation from multiple droplets DP in accordance with some embodiments of the disclosure. The EUV lithography system 800 includes an analyzer module 830 and a controller 840 coupled to each other. The analyzer module 830 receives the images of the droplets DP through the DDM 820 that is consistent with one of the DDMS 620A, 620B, 620C, or 620D and determines, e.g., calculates, a timing of the laser generator 310 to generate the later pulses that generate the pattern of bright and dark fringes 520 and 535 based on the location of the multiple droplets DP. The analyzer module 830 may command the laser generator 310 through the controller 840. In some embodiments, based on the a timing of the laser generator 310, the analyzer module 830 commands the droplet generator 115 through the controller 840 to eject the droplets DP. In some embodiments, the analyzer module 830 commands the DIM 810 that is consistent with one of the DIMS 610A, 610B, 610C, or 610D to irradiate by non-ionizing light the zone of excitation ZE to illuminate the droplets DP.

FIGS. 9A and 9B illustrate flow diagrams of an exemplary processes for generating EUV radiation from multiple droplets in accordance with some embodiments of the disclosure. The process 900 of FIG. 9A may be performed by EUV lithography system of FIGS. 1 and 8 and use the EUV radiation sources of 6A, 6B, 7A, 7B. In some embodiments, the process 900 or a portion of the process 900 is performed and/or is controlled by the computer system 1000 described below with respect to FIGS. 10A and 10B. The method includes the operation S910 of generating two or more fringes by a laser beam passing through a pair of slits. As shown in FIG. 7A, the ionizing laser beams 506 pass through the pair of slits 522 and the pattern of bright and dark fringes 520 is generated. In operation S920, two or more droplets DP are irradiated by the two or more bright fringes 527 of the pattern of bright and dark fringes 520 to produce EUV radiation. As shown in FIG. 7A, the droplets DP pass through the bright fringes 527 and are irradiated by the ionizing laser beam 506 and the EUV light rays 24 (EUV radiation) are produced. In operation S930, the EUV radiation produced from the two or more droplets DP is collected and directed by a collector mirror. As shown in FIG. 7A, the EUV light rays 24 are collected and directed by the collector mirror 110 to the opening 148 of FIG. 7A. In operation S940, the collected and directed EUV radiation is used in a lithographic system, e.g., the system of FIG. 2, to image a photo mask, e.g., the reflective mask 205c of FIG. 2 on a substrate, e.g., the target semiconductor substrate 210.

The process 950 of FIG. 9B may be performed by EUV lithography system of FIGS. 1 and 8 and use the EUV radiation sources of 6B and 7B. In some embodiments, the process 950 or a portion of the process 950 is performed and/or is controlled by the computer system 1000 described below with respect to FIGS. 10A and 10B. The method includes the operation S952 of generating two or more first fringes by an ionizing laser beam passing through a first pair

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of slits. As shown in FIG. 7B, the ionizing laser beams 506 pass through the pair of slits 522 and the two or more first bright fringes 527 of the pattern of bright and dark fringes 520 are generated in the zone of excitation ZE. In operation 954, two or more second fringes are generated by a non-ionizing laser beam passing through a second pair of slits. As shown in FIG. 7B, the non-ionizing laser beams 606 pass through the pair of slits 530 and the two or more second bright fringes 537 of the pattern of bright and dark fringes 535 are generated in the zone of excitation ZE.

In operation S956, two or more droplets DP are irradiated by the two or more second bright fringes 537 of the pattern of bright and dark fringes 535 to pre-heat the two or more droplets DP to generate two or more pancaked-shaped droplets 27. In operation S958, the two or pancaked-shaped droplets 27 are irradiated by the two or more first bright fringes 527 of the pattern of bright and dark fringes 520 to produce EUV radiation. In operation S960, the EUV radiation produced from the two or more pancaked-shape droplets 27 is collected and directed by a collector mirror. As shown in FIG. 7A, the EUV light rays 24 are collected and directed by the collector mirror 110 to the opening 148 of FIG. 7B. In operation S962, the collected and directed EUV radiation is used in a lithographic system, e.g., the system of FIG. 2, to image a photo mask, e.g., the reflective mask 205c of FIG. 2 on a substrate, e.g., target semiconductor substrate 210.

FIGS. 10A and 10B illustrate a computer system for an apparatus for generating EUV radiation from multiple droplets in accordance with some embodiments of the disclosure. FIG. 10A is a schematic view of a computer system 1000 that executes the process for generating EUV radiation from multiple droplets according to one or more embodiments as described above. All of or a part of the processes, method and/or operations of the foregoing embodiments can be realized using computer hardware and computer programs executed thereon. In some embodiments, the computer system 1000 provides the functionality of the controller 840, the analyzer module 830 of FIG. 8. In FIG. 10A, a computer system 1000 is provided with a computer 1001 including an optical disk read only memory (e.g., CD-ROM or DVD-ROM) drive 1005 and a magnetic disk drive 1006, a keyboard 1002, a mouse 1003, and a monitor 1004.

FIG. 10B is a diagram showing an internal configuration of the computer system 1000. In FIG. 10B, the computer 1001 is provided with, in addition to the optical disk drive 1005 and the magnetic disk drive 1006, one or more processors 1011, such as a micro-processor unit (MPU), a ROM 1012 in which a program such as a boot up program is stored, a random access memory (RAM) 1013 that is connected to the processors 1011 and in which a command of an application program is temporarily stored and a temporary storage area is provided, a hard disk 1014 in which an application program, a system program, and data are stored, and a bus 1015 that connects the processors 1011, the ROM 1012, and the like. Note that the computer 1001 may include a network card (not shown) for providing a connection to a LAN.

The program for causing the computer system 1000 to execute the process for generating EUV radiation from multiple droplets in the foregoing embodiments may be stored in an optical disk 1021 or a magnetic disk 1022, which are inserted into the optical disk drive 1005 or the magnetic disk drive 1006, and transmitted to the hard disk 1014. Alternatively, the program may be transmitted via a network (not shown) to the computer 1001 and stored in the hard disk 1014. At the time of execution, the program is

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loaded into the RAM 1013. The program may be loaded from the optical disk 1021 or the magnetic disk 1022, or directly from a network. The program does not necessarily have to include, for example, an operating system (OS) or a third party program to cause the computer 1001 to execute the process for manufacturing the lithographic mask of a semiconductor device in the foregoing embodiments. The program may only include a command portion to call an appropriate function (module) in a controlled mode and obtain desired results.

According to some embodiments of the present disclosure, a method for generating EUV radiation includes simultaneously irradiating two or more target droplets with a first laser beam in an EUV radiation source apparatus to produce EUV radiation. The method also includes collecting and directing the EUV radiation produced from the two or more target droplets by an imaging mirror. In an embodiment, the method further includes configuring a droplet generator to release the two or more target droplets in first parallel lines in a first plane. In an embodiment, the method further includes dividing the first laser beam into two portions, interfering the two portions of the first laser beam to produce two or more fringes, and irradiating each one of the two or more target droplets with one of the two or more fringes. In an embodiment, the method further includes configuring the two portions of the first laser beam to pass through a pair of slits to generate the two or more fringes. In an embodiment, the method further includes generating the two or more fringes in the first plane. The two or more fringes are in parallel with the first parallel lines, and configuring each one of the two or more target droplets to simultaneously pass through a separate fringe of the two or more fringes. In an embodiment, the first plane is perpendicular to a direction of the first laser beam and the method further includes using the collected and directed EUV radiation in a lithographic system to image a photo mask on a substrate. In an embodiment, the method further includes dividing the first laser beam into three or more portions, interfering the three or more portions of the first laser beam to produce the two or more fringes, and irradiating each one of the two or more target droplets with one of the two or more fringes. In an embodiment, the method further includes generating a plasma plume by irradiating the two or more target droplets with the two or more fringes of the first laser beam. The EUV radiation is generated by the plasma plume. In an embodiment, the method further includes configuring the two portions of the first laser beam to pass through different path lengths, and generating a shift of a location of the two or more fringes by the different path lengths. In an embodiment, the method further includes simultaneously pre-heating the two or more target droplets with a second laser beam in the EUV radiation source, and generating a pancake-shaped droplet by the pre-heating. The pre-heating is before the irradiating the two or more target droplets with a first laser beam.

According to some embodiments of the present disclosure, a method of for generating EUV radiation includes generating two or more fringes by a first laser beam passing through a first pair of slits. The method includes irradiating two or more target droplets by the two or more fringes to produce EUV radiation. Each one of the two or more target droplets simultaneously pass one of the two or more fringes. The method also includes collecting and directing the EUV radiation produced from the two or more target droplets by an imaging mirror. In an embodiment, the method further includes dividing the first laser beam into two portions, and interfering the two portions of the first laser beam to produce

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the two or more fringes. In an embodiment, the method further includes generating a plasma plume by irradiating the two or more target droplets with the two or more fringes of the first laser beam. The EUV radiation is generated by the plasma plume. In an embodiment, the method further includes configuring a droplet generator to simultaneously release the two or more target droplets.

According to some embodiments of the present disclosure, An EUV radiation source includes a first laser source configured to generate a first laser beam and a collector mirror having two or more slits. The radiation source includes a droplet generator configured to simultaneously release two or more target droplets in first parallel lines in a first plane. The radiation source also includes an optical system configured to divide the first laser beam into two or more portions and to pass the two or more portions of the first laser beam through the two or more slits to generate two or more first fringes in parallel with a path of the two or more target droplets. The radiation source includes a controller coupled to and controlling the first laser source and the droplet generator. The controller is configured to fire a laser pulse of the first laser source to irradiate the simultaneously released two or more target droplets by the two or more first fringes. Each first fringe irradiates one droplet. In an embodiment, the optical system of the EUV radiation source includes a lens configured to generate a collimated beam of the first laser beam. The collimated beam is configured to impinge on the two or more slits. In an embodiment, the slits have a rectangular shape. In an embodiment, the EUV radiation source further includes a second laser source configured to generate a second pre-heat laser beam. The optical system is further configured to divide the second pre-heat laser beam into two or more portions and to pass the two or more portions of the second pre-heat laser beam through the two or more slits to generate two or more second fringes in parallel with a path of the two or more target droplets. In an embodiment, the optical system is configured to generate a first number of first fringes that is equal to a second number of second fringes and the first number of first fringes and the second number of second fringes are between 2 and 8, generate the two or more first fringes at a zone of excitation, and generate the two or more second fringes in a path of the two or more target droplets from the droplet generator to the zone of excitation and before the zone of excitation. In an embodiment, the EUV radiation source further includes a droplet illumination module configured to irradiate the zone of excitation with non-ionizing light to illuminate the two or more target droplets, and a droplet detection module configured to capture images from the zone of excitation to determine a velocity of the two or more target droplets and to determine, based on the velocity, a timing of turning the first and second laser sources on and off.

In some embodiments, implementing the processes and methods mentioned above, increases the efficiency of generating the EUV radiation by simultaneously irradiating multiple droplets.

The foregoing outlines features of several embodiments or examples so that those skilled in the art may better understand the aspects of the present disclosure. Those skilled in the art should appreciate that they may readily use the present disclosure as a basis for designing or modifying other processes and structures for carrying out the same purposes and/or achieving the same advantages of the embodiments or examples introduced herein. Those skilled in the art should also realize that such equivalent constructions do not depart from the spirit and scope of the present

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disclosure, and that they may make various changes, substitutions, and alterations herein without departing from the spirit and scope of the present disclosure.

What is claimed is:

1. A method of manufacturing an integrated circuit, comprising:

generating an ionizing laser beam;

dividing the laser beam into a first laser beam and a second laser beam;

passing the first and second laser beams respectively through a first slit and a second slit formed in a collector mirror to form a first pattern of bright fringes in a zone of excitation;

generating a plurality of droplets to simultaneously drop through the first pattern of bright fringes to generate extreme ultraviolet (EUV) light, the collector mirror directing the EUV light to a reflective mask to generate a patterned beam of light; and

directing the patterned beam of light onto a photoresist formed on a semiconductor substrate.

2. The method of claim 1, wherein the first and second laser beams are in phase.

3. The method of claim 1, wherein the first and second laser beams are 180° out of phase.

4. The method of claim 1, wherein a distance between centers of the first and second slits ranges from 0.5 mm to 10 mm.

5. The method of claim 1, further comprising passing a third laser beam and fourth laser beam respectively through a third slit and a fourth slit formed in the collector mirror to generate a second pattern of bright fringes.

6. The method of claim 5, wherein a distance between centers of the first and second slits is different from a distance between centers of the third and fourth slits.

7. The method of claim 5, further comprising passing the plurality of droplets through the second pattern of bright fringes before passing the plurality of droplets through the first pattern of bright fringes.

8. The method of claim 5, wherein the first pattern of bright fringes and the second pattern of bright fringes are within an area having dimensions of 500 μm by 500 μm in the zone of excitation.

9. A method of manufacturing an integrated circuit, comprising:

processing an ionizing laser beam into a plurality of bright fringes in a zone of excitation;

simultaneously passing a plurality of droplets through the plurality of bright fringes to generate extreme ultraviolet (EUV) light;

focusing and directing the generated EUV light onto a reflective mask to generate a patterned beam of light; and

exposing a photoresist formed on a semiconductor substrate to the patterned beam of light.

10. The method of claim 9, wherein the processing of the ionizing laser beam comprises passing the laser beam through a beam splitter to generate a pair of laser beams, and passing the pair of laser beams respectively through a pair of slits formed in a mirror to generate the plurality of bright fringes.

11. The method of claim 9, further comprising generating an additional plurality of bright fringes.

12. The method of claim 11, further comprising passing the plurality of droplets through the additional plurality of bright fringes to preheat the droplets before passing the droplets through the plurality of bright fringes to generate the EUV light.

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13. A method of manufacturing an integrated circuit, comprising:

passing a first ionizing laser beam through a first slit and passing a second ionizing laser beam through a second slit to collectively generate a first plurality of bright fringes in a zone of excitation;

simultaneously passing separate droplets through the first plurality of bright fringes to generate extreme ultraviolet (EUV) light;

reflecting the EUV light off a reflective mask to generate a patterned beam of light; and

exposing a photoresist formed on a semiconductor substrate to the patterned beam of light.

14. The method of claim 13, further comprising passing a first non-ionizing laser beam through a third slit and passing a second non-ionizing laser beam through a fourth slit to collectively generate a second plurality of bright fringes.

15. The method of claim 14, further comprising simultaneously passing the separate droplets through the second

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plurality of bright fringes before passing the passing the separate droplets through the first plurality of bright fringes.

16. The method of claim 15, wherein the second plurality of bright fringes pre-heats the separate droplets before the separate droplets pass through the first plurality of bright fringes.

17. The method of claim 16, wherein the droplets have a diameter ranging from 10 μm to 22 μm before passing through the second plurality of bright fringes.

18. The method of claim 14, wherein the first and second non-ionizing laser beams are pulsed laser beams.

19. The method of claim 13, wherein the droplets pass through the first plurality of bright fringes at a speed ranging from 10 m/s to 100 m/s.

20. The method of claim 13, wherein the first and second ionizing laser beams are pulsed laser beams.

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